# Star Vetter - confirmed eclipsing binary candidates

| TIC | Period (d) | Depth | BLS SNR | Pixel SNR | Offset | Tier |
|---|---|---|---|---|---|---|
| 437150434 | 5.2853 | 15.003% | 1081 | 85.33 | 1" | Clean |
| 237115819 | 14.7921 | 8.883% | 507 | 27.82 | 4" | Clean |
| 66732476 | 3.1087 | 9.814% | 370 | 26.93 | 8" | Clean |
| 29495157 | 5.1474 | 8.579% | 333 | 52.62 | 5" | Alias |
| 119459681 | 5.9442 | 15.977% | 311 | 68.15 | 2" | Clean |
| 411434067 | 1.3536 | 4.289% | 193 | 33.70 | 4" | Alias |
| 330901906 | 3.7321 | 0.498% | 60 | 87.24 | 4" | Thin margin |
| 303470667 | 5.4319 | 2.220% | 57 | 5.02 | 9" | Marginal |
| 273158289 | 1.1519 | 1.255% | 41 | 17.24 | 2" | Thin margin |
| 111149941 | 5.2401 | 0.045% | 10 | 9.73 | 21" | Ambiguous |
| 251898081 | 27.3089 | 18.630% | 5535 | 464.16 | 3" | Alias |
| 467214801 | 0.7609 | 0.950% | 2134 | 21.36 | 0" | Alias |
| 399770041 | 1.5802 | 4.291% | 489 | 36.92 | 1" | Clean |
| 349701853 | 30.3427 | 28.445% | 343 | 51.05 | 6" | Clean |
| 233459038 | 25.3384 | 8.315% | 237 | 84.46 | 5" | Alias |
| 281310348 | 1.7702 | 7.273% | 180 | 57.87 | 2" | Clean |
| 318887646 | 14.2817 | 3.605% | 150 | 6.94 | 13" | Clean |
| 444465823 | 18.5133 | 3.476% | 147 | 25.18 | 1" | Alias |
| 380943251 | 0.7209 | 7.667% | 101 | 15.26 | 3" | Clean |
| 66716482 | 8.0283 | 3.329% | 65 | 21.51 | 7" | Clean |
| 309656476 | 28.9204 | 0.050% | 25 | 13.80 | 14" | Clean |
| 229494833 | 15.3839 | 0.037% | 15 | 5.19 | 12" | Alias |
| 44533527 | 14.5952 | 0.101% | 12 | 7.33 | 16" | Clean |
| 220423668 | 3.6511 | 4.525% | 755 | 17.67 | 6" | Alias |
| 365998904 | 1.2494 | 2.398% | 375 | 15.52 | 3" | Clean |
| 233452636 | 1.9063 | 5.467% | 202 | 8.94 | 2" | Thin margin |
| 437171483 | 1.4735 | 4.947% | 923 | 73.37 | 3" | Clean |
| 411457490 | 2.4324 | 5.531% | 431 | 58.00 | 1" | Alias |
| 295373509 | 5.4219 | 8.026% | 247 | 22.12 | 2" | Clean |
| 374158907 | 14.6563 | 4.974% | 179 | 21.40 | 2" | Alias |

# TIC 40208362

Eclipsing binary candidate dossier - Star Vetter

**CLEAN**

```
BLS EPHEMERIS
  Period        0.690816 d
  Epoch (t0)    1711.5018 BTJD
  Duration      0.0400 d
  Depth         1.629 %  (16,287 ppm)
  SNR           240.1

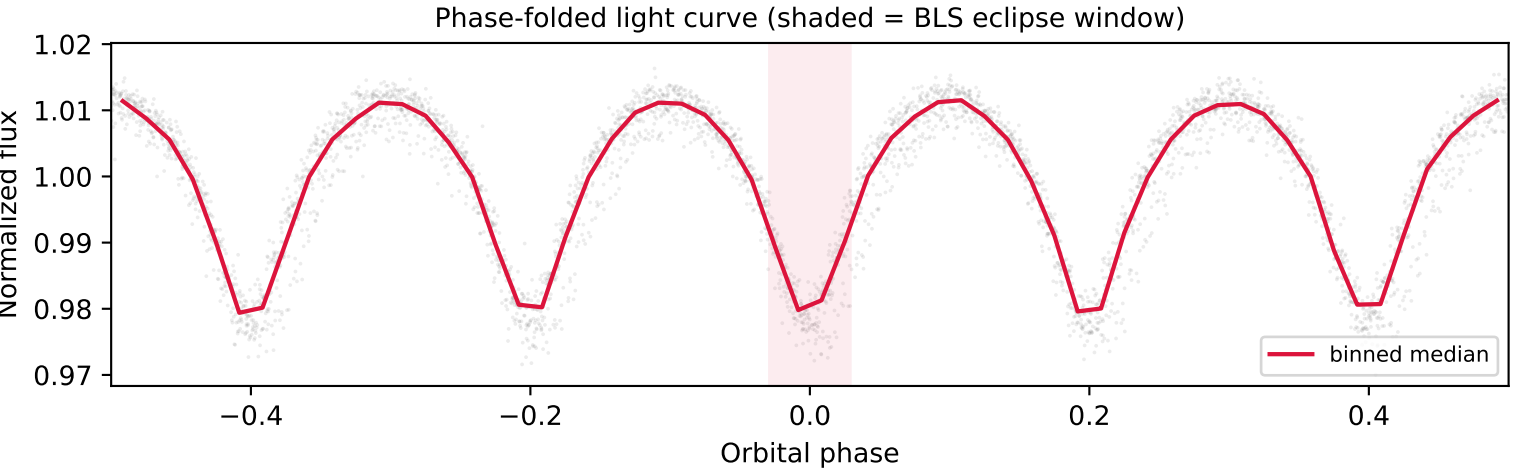
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      36  (gate >= 3)
  Robust depth ratio     1.27  (gate >= 0.50)
  Robust depth sigma     27.0  (gate >= 7.0)
  Odd/even z-score       1.22  (consistent)
  Secondary eclipse      not detected
```

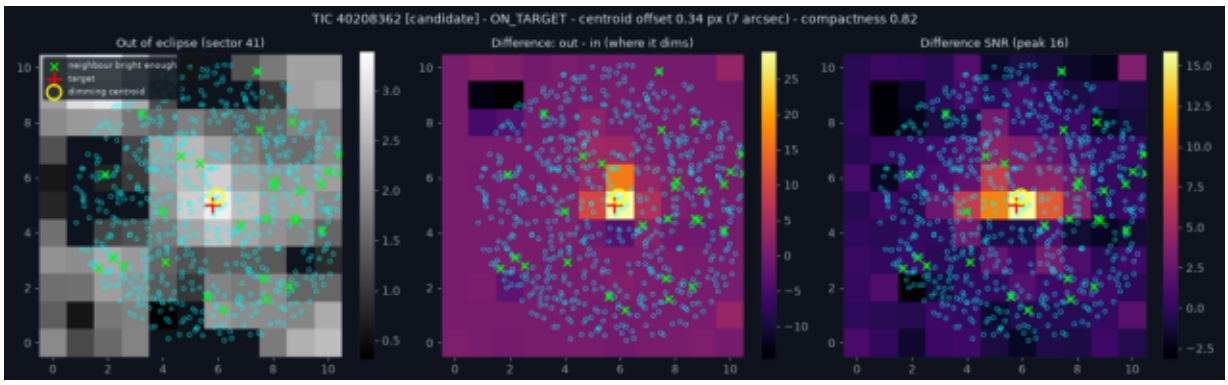
```
CATALOG CROSS-MATCH (novelty)

  VSX                                  no match
  ASAS-SN                              no match
  Gaia eclipsing-binary table          no match
  Gaia variable classification         no match
  TESS EB catalog (Prsa+2022)          no match
  ExoFOP TOI list                      no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 41    light-curve dip in this sector: 19.8 sigma    t0 refined by 0.2000 phase
  Difference-image peak SNR   15.91   (gate >= 5.0)      Compactness   0.82   (gate >= 0.50)
  Dimming centroid offset     0.34 px (7 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 40208348
    1.33 px away, dTmag +4.32 - farther from the dimming than the target
```




No caveats - clean detection on every check.

# TIC 295254637

Eclipsing binary candidate dossier - Star Vetter

**PERIOD ALIAS**

```
BLS EPHEMERIS
  Period       2.664955 d  (ALIASED, see below)
    -> corrected period: 5.329910 d
  Epoch (t0)   1626.9366 BTJD
  Duration     0.1200 d
  Depth        3.867 %  (38,674 ppm)
  SNR          176.9

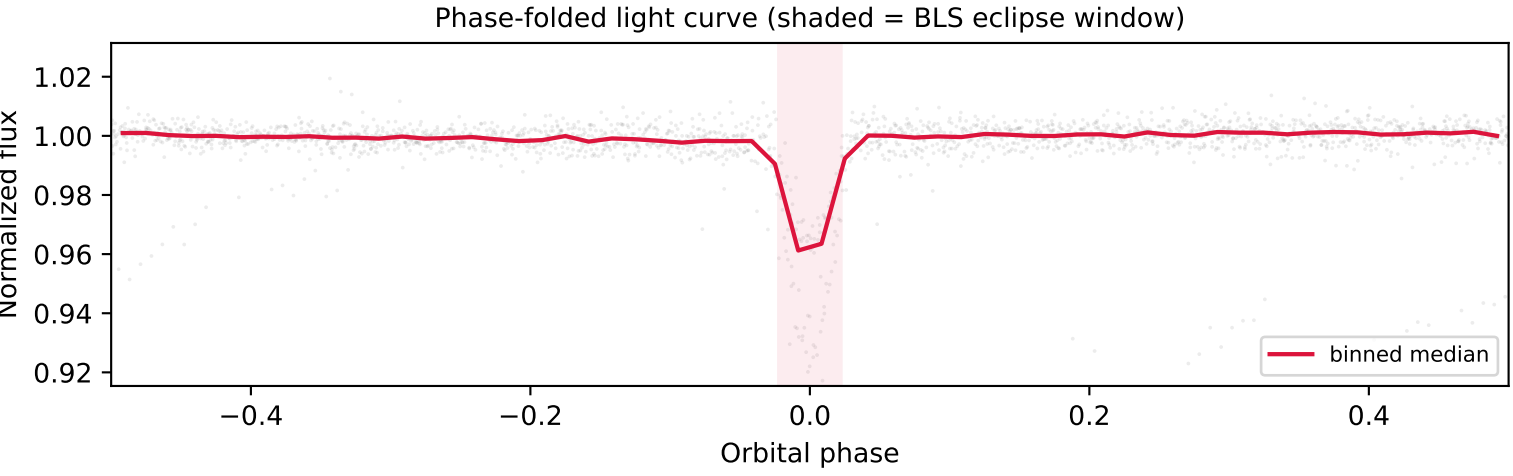
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     17  (gate >= 3)
  Robust depth ratio    0.87  (gate >= 0.50)
  Robust depth sigma    74.0  (gate >= 7.0)
  Odd/even z-score      15.37  (ALIAS FLAGGED)
  Secondary eclipse     not detected
```

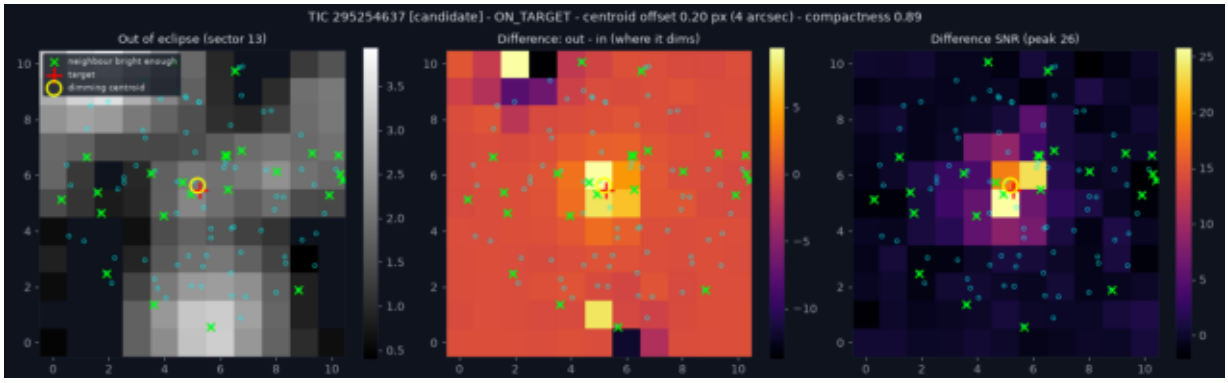
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 13    light-curve dip in this sector: 57.0 sigma    t0 refined by -0.0025 phase
  Difference-image peak SNR   25.85   (gate >= 5.0)      Compactness   0.89   (gate >= 0.50)
  Dimming centroid offset     0.20 px (4 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 295254639
    0.40 px away, dTmag +0.13 - farther from the dimming than the target
```




Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 15.37 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 5.329910 d, in any write-up.

# TIC 468210292

Eclipsing binary candidate dossier - Star Vetter

**PERIOD ALIAS**

```
BLS EPHEMERIS
  Period       1.445077 d  (ALIASED, see below)
    -> corrected period: 2.890154 d
  Epoch (t0)   1569.6212 BTJD
  Duration     0.1200 d
  Depth        7.541 %  (75,412 ppm)
  SNR          140.9

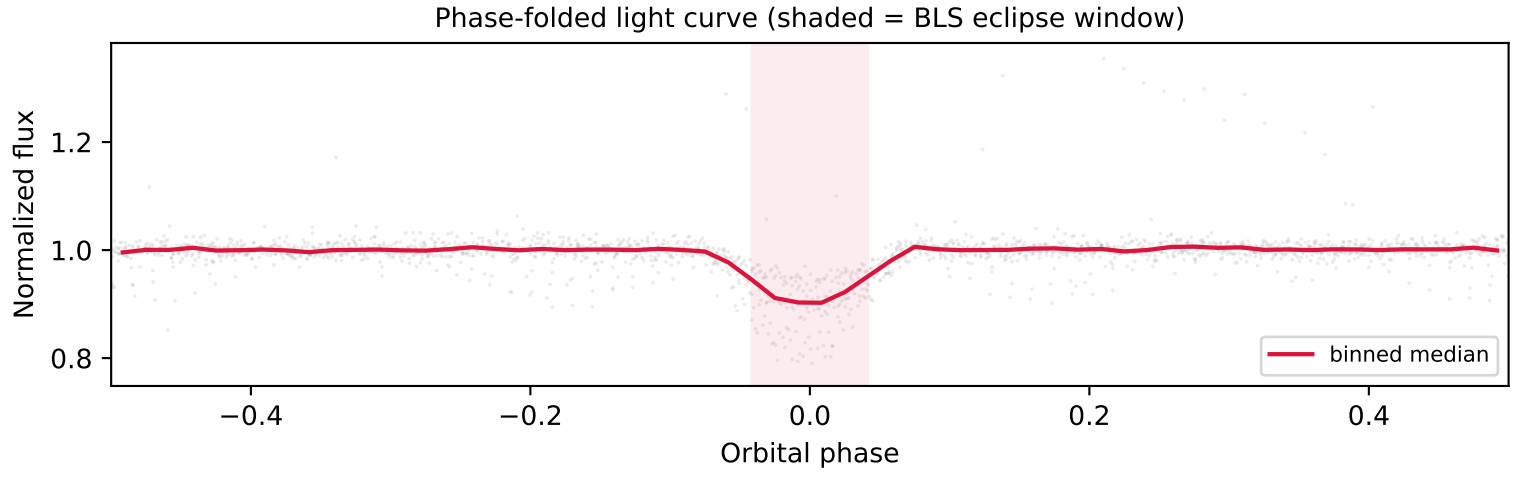
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     31  (gate >= 3)
  Robust depth ratio    1.13  (gate >= 0.50)
  Robust depth sigma    77.6  (gate >= 7.0)
  Odd/even z-score      9.36  (ALIAS FLAGGED)
  Secondary eclipse     not detected
```

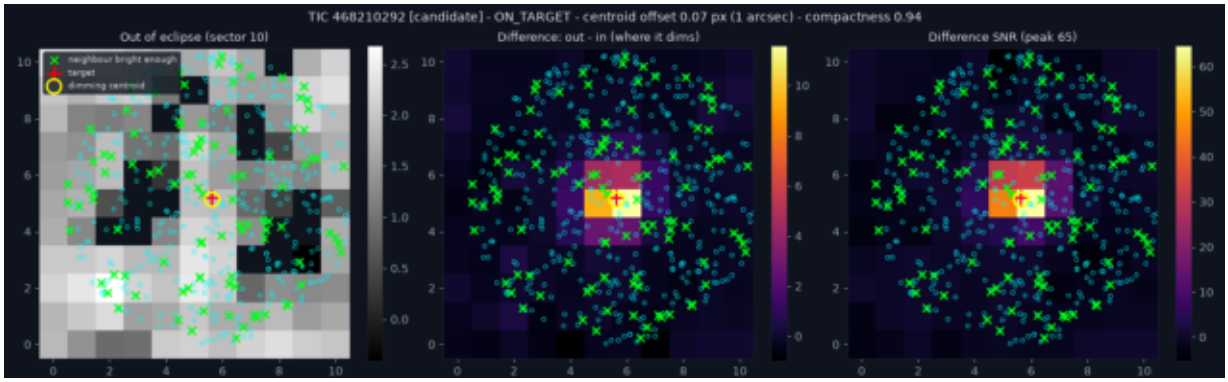
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 10    light-curve dip in this sector: 57.7 sigma    t0 refined by -0.0050 phase
  Difference-image peak SNR   64.75   (gate >= 5.0)      Compactness   0.94   (gate >= 0.50)
  Dimming centroid offset     0.07 px (1 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 468210271
    0.40 px away, dTmag +1.68 - farther from the dimming than the target
```




Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 9.36 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 2.890154 d, in any write-up.

# TIC 258151016

Eclipsing binary candidate dossier - Star Vetter

**CLEAN**

```
BLS EPHEMERIS
  Period        0.894047 d
  Epoch (t0)    1518.3672 BTJD
  Duration      0.1200 d
  Depth         3.871 %  (38,706 ppm)
  SNR           114.5

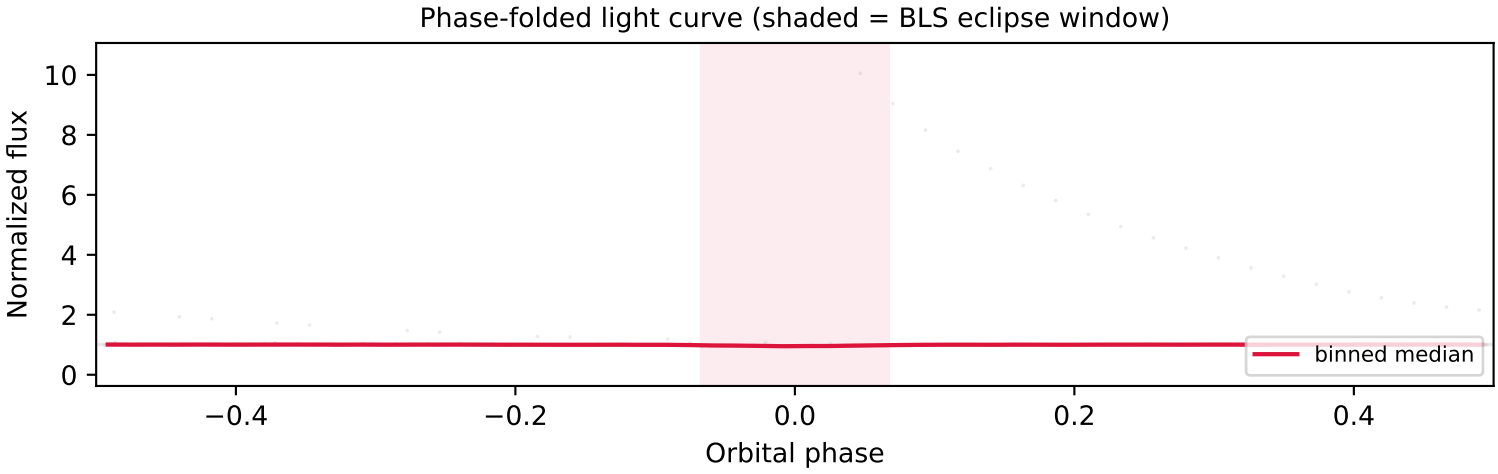
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed      19  (gate >= 3)
  Robust depth ratio     0.99  (gate >= 0.50)
  Robust depth sigma     59.4  (gate >= 7.0)
  Odd/even z-score       0.24  (consistent)
  Secondary eclipse      not detected
```

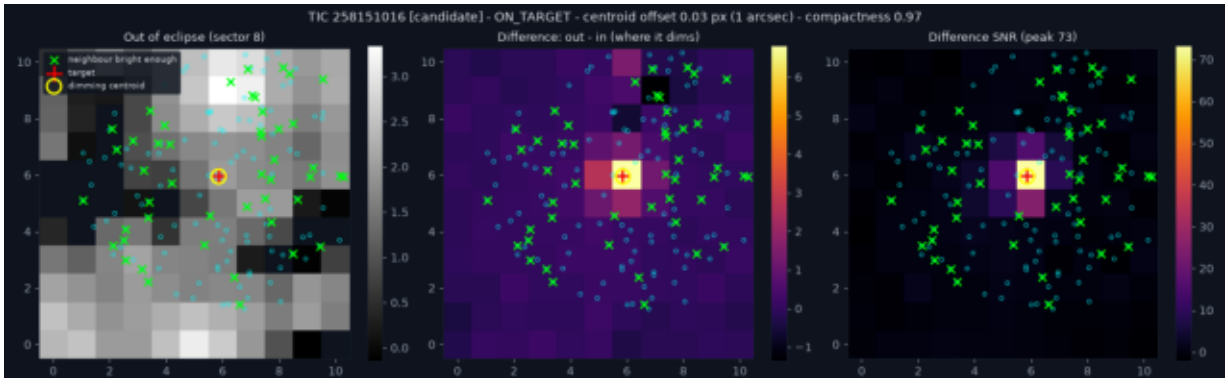
```
CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
```

```
PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 8    light-curve dip in this sector: 59.5 sigma    t0 refined by 0.0025 phase
  Difference-image peak SNR   73.44   (gate >= 5.0)      Compactness   0.97   (gate >= 0.50)
  Dimming centroid offset     0.03 px (1 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 258151031
    1.41 px away, dTmag +2.12 - farther from the dimming than the target
```




No caveats - clean detection on every check.